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EDUCATION

Rutgers University–New Brunswick, School of Engineering | B.S. Mechanical Engineering Class of 2030

HONORS

Sir Isaac Newton Physics Exam: 9th internationally of 3,800+ (2025) | **USAPhO:** Distinction, top 400 of 6,000 (2025)
NASA App Development Challenge: National Winner, 200+ teams (2025) | **MIT Beaver Works:** 4th of 100 teams (2025)
NJ State Science Fair: 2nd Place, Computer Science, flood model (2026); Air Force Award, drone-swarm RL (2025)

PUBLICATION

Beyond the Haystack: Sensitivity to Context in Legal Reference Recall. Third author. NLLP Workshop at EMNLP 2025.
• Evaluation framework for long-context LLM recall on three naturalistic legal-reference tasks built from U.S. court-opinion briefings; exposed behavior missed by synthetic benchmarks such as Needle in a Haystack.

RESEARCH & ENGINEERING PROJECTS

BLUEBIRD Survey Drone *Designed and simulated; prototype next* 2026

- Designed a 1.8 m fixed-wing drone for conservation and agriculture in FreeCAD, with a printable center body and removable foam wings; produced full CAD and an 86-page engineering report.
- Developed Android RGB/thermal perception and mission logic with independent ArduPilot control and failsafes; refined and evaluated the design using SU2 CFD, CalculiX FEA and ArduPilot software-in-the-loop tests before fabrication.

Physics-Informed Flood Forecasting Model *Science fair research* 2025–2026

- Built a PyTorch graph transformer for flood forecasting, with the 1D Saint-Venant continuity equation in its loss to penalize predictions that violate conservation of mass.
- In project benchmarks against a finite-difference reference, reached 117× faster inference with 97% mass-conservation adherence.

Cable-Driven Soft Robotic Arm *Independent; built and bench-tested* 2024–2026

- Built and bench-tested a three-cable TPU continuum arm for 3D bending; evaluated compliance and response.
- Added camera-based hand-pose control and a target-trajectory visualizer; developing a VLM + heuristic solver controller.

Kestrel Mini R4 Bicopter *Build in progress; not yet flown* 2026

- Designed a FreeCAD bicopter with twin lift motors, independent tilt servos, 40 printed parts and a 42-route wire schedule; building the first physical prototype.
- Estimated 665 g and 1.82 thrust-to-weight at 12 V with 15% installation loss; thrust, servo, print-strength and flight tests pending.

SA-2 Samson Tilt-Nacelle VTOL *CAD prototype; build planned after Kestrel* 2026

- Modeled a 1:15 scale RC VTOL in FreeCAD with twin tilting nacelles, coaxial 12 in rotors and ArduPilot QuadPlane, plus a bicopter variant. Wrote a staged structural, propulsion, tilt and hover test plan; checked mass and geometry with Python.

Axiom: Offline Math Notebook *macOS desktop application* 2026

- Built an offline notebook with typeset equations, matrices, 2D/3D graphs, multivariable calculus, a reference-PDF pane and PDF export.
- Added a 2D/3D truss solver that checks determinacy, solves forces and reactions, and grades answers; notarization in progress.

Active Aerodynamics RC Testbed *Independent; in development* 2023–2026

- Built a 1/8-scale RC car with a ground-effect downforce fan and drag-reduction system; ECU runs on a repurposed smartphone with on-device perception.

NASA App Development Challenge, Team Luna Vista *National Winner* 2025

- Modeled the Artemis II flight path in Unreal Engine 5 using orbital mechanics and C++ shaders; optimized antenna scheduling for the visualization.

SKILLS & TRAINING

CAD & simulation: FreeCAD; SU2 and OpenFOAM (CFD), CalculiX (FEA), ArduPilot SITL

Hardware: 3D-printed TPU and structural parts, cable actuation, RC electronics and wiring layouts

Software & modeling: Python, C++, Unreal Engine 5, physics-informed ML, reinforcement learning, model evaluation

Safety: OSHA-10